

# Predictive UPF steering: causal control, national-scale evidence

## A 45-minute technical showcase

**Standards-grounded and statistically verified synthetic modeling at national scale, but not yet calibrated to C-DOT production traffic.**

LIVE LOOP

**3 UPFs**

Causal, offline, new sessions only

FROZEN SCALE

**24 UPFs**

96 groups · 16M aggregate UEs

# Start with the evidence boundary—not the promise

- The 3-UPF runtime is causal and steers only future sessions.
- The guided campaign improves mean-pair UL overload area by 10.52%.
- The 30-pair national MA6 MPC result passes its campaign gate with visible fault tails.
- The 14-day forecaster is more accurate but its MPC pairing fails the control gate.
- Continuous oracle relaxations show 49–69% causal/scheduled and 100% clairvoyant headroom.
- Standards-grounded and statistically verified synthetic modeling at national scale, but not yet calibrated to C-DOT production traffic.

LIVE

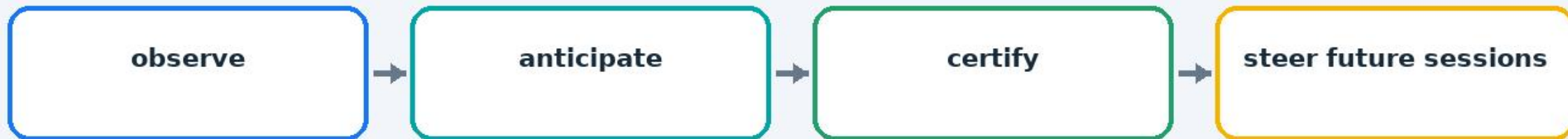
MEASURED - SYNTHETIC

MODELED - PROJECTION

EXTERNAL - PENDING

# Why predict? Long-lived sessions make late reaction expensive

- Current utilization is not the future state: today's admissions persist into later fault and surge windows.
- The actuator can change weights for new sessions; it cannot migrate established sessions.
- Forecasting matters only when optimization, certification and actuation convert information into safer placement.



# National-scale synthetic topology, bounded into tractable cohorts

AGGREGATE UE POPULATION

**16M**

exactly conserved

UPFS / ZONES

**24 / 8**

edge · regional · central

TRAFFIC GROUPS

**96**

12 services × 8 zones

TELEMETRY CADENCE

**30 s**

causal event-time ticks

**Aggregate UE cohorts are modeled; persistent subscriber identities are not.**



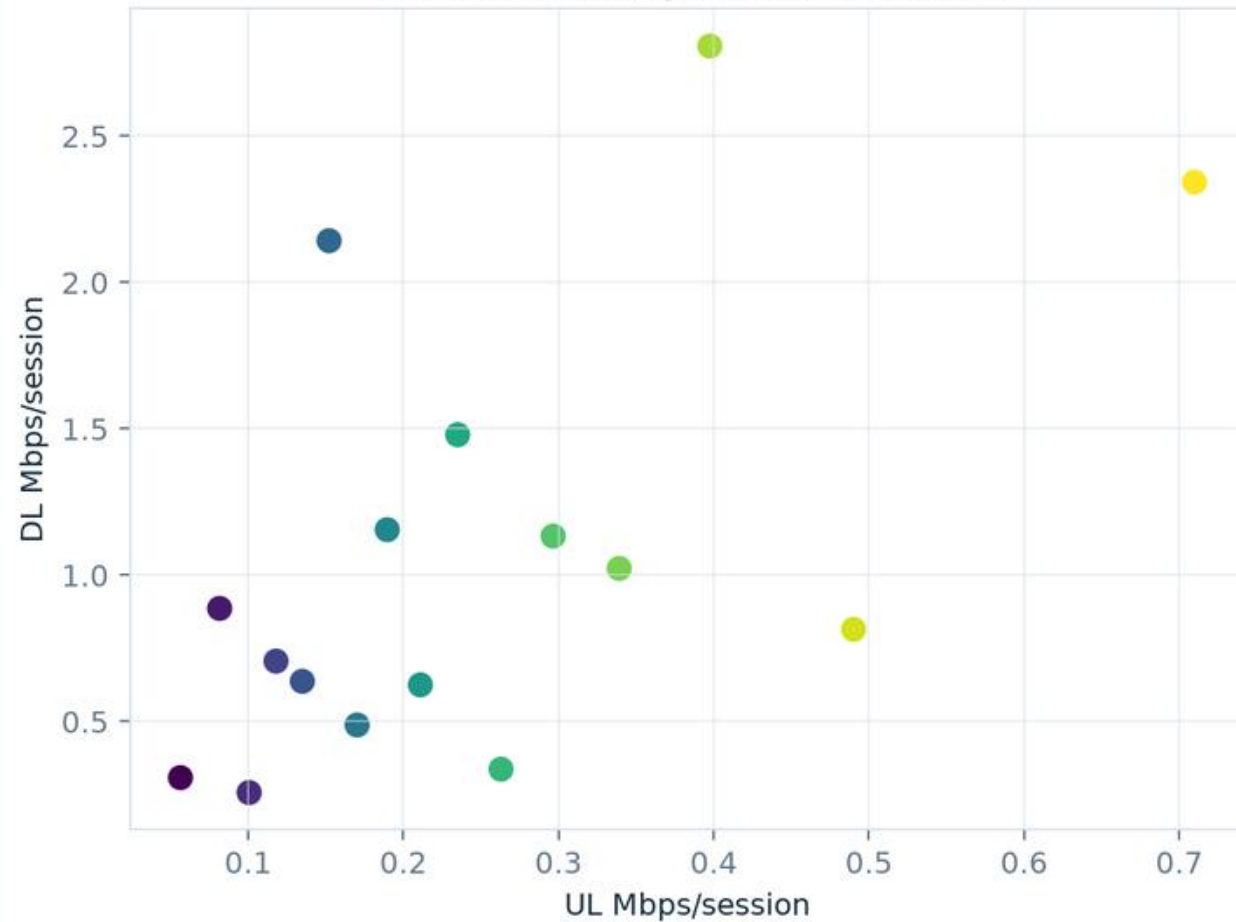
# Realism scorecard: strong structure, explicit calibration gap



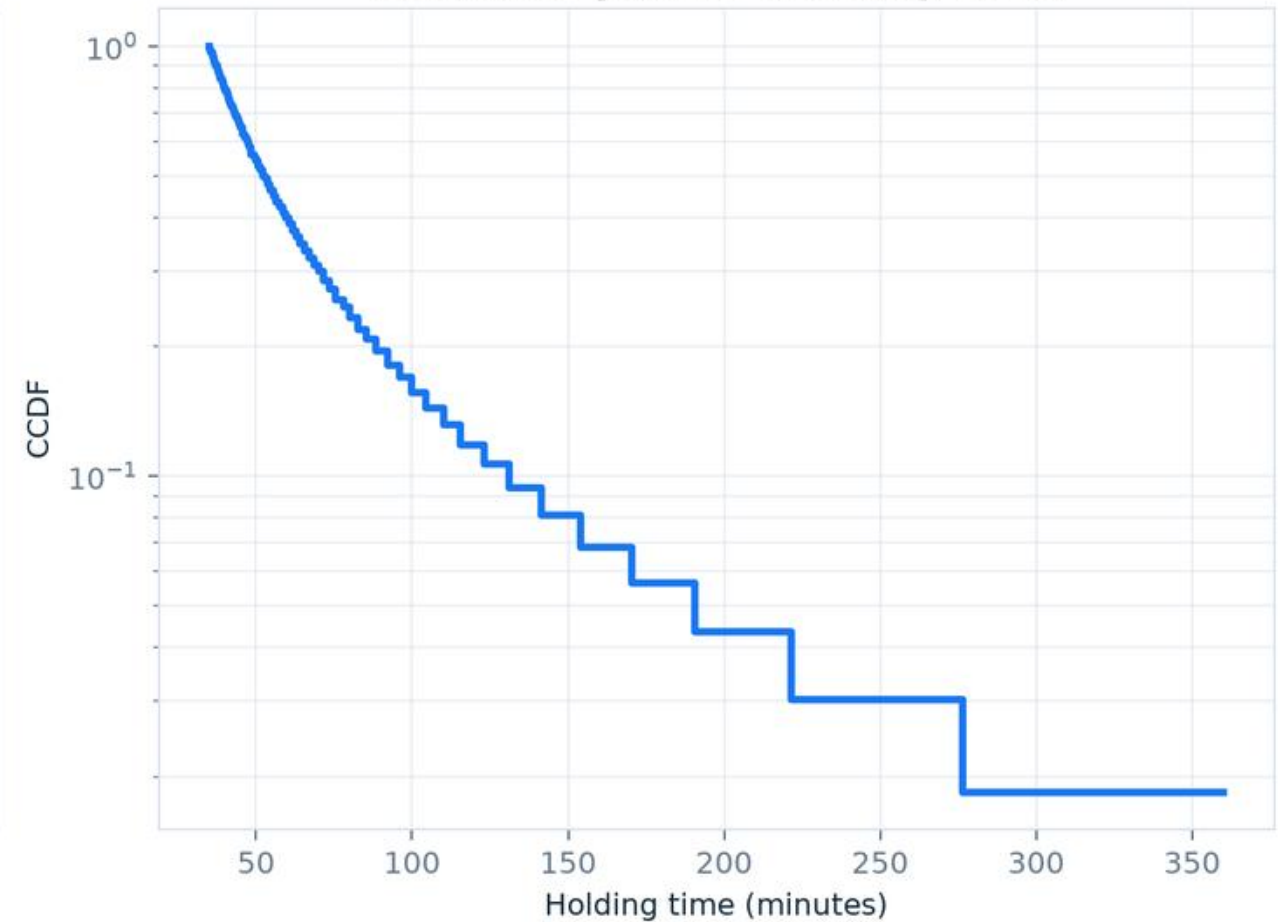
**Standards-grounded and statistically verified synthetic modeling at national scale, but not yet calibrated to C-DOT production traffic.**

# Distribution fidelity passes all configured acceptance checks

## 16 bounded joint rate bins

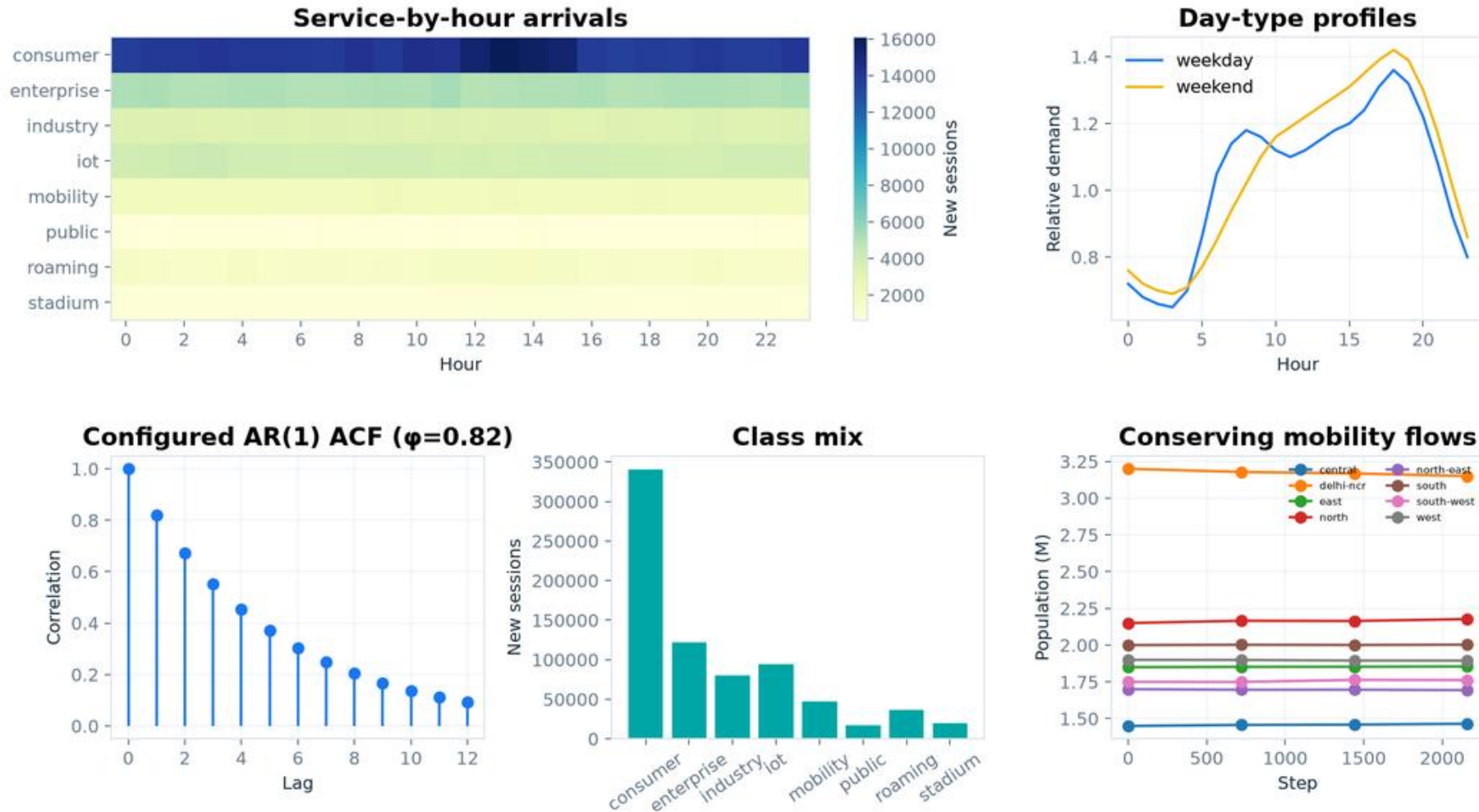


## Bounded pareto holding time



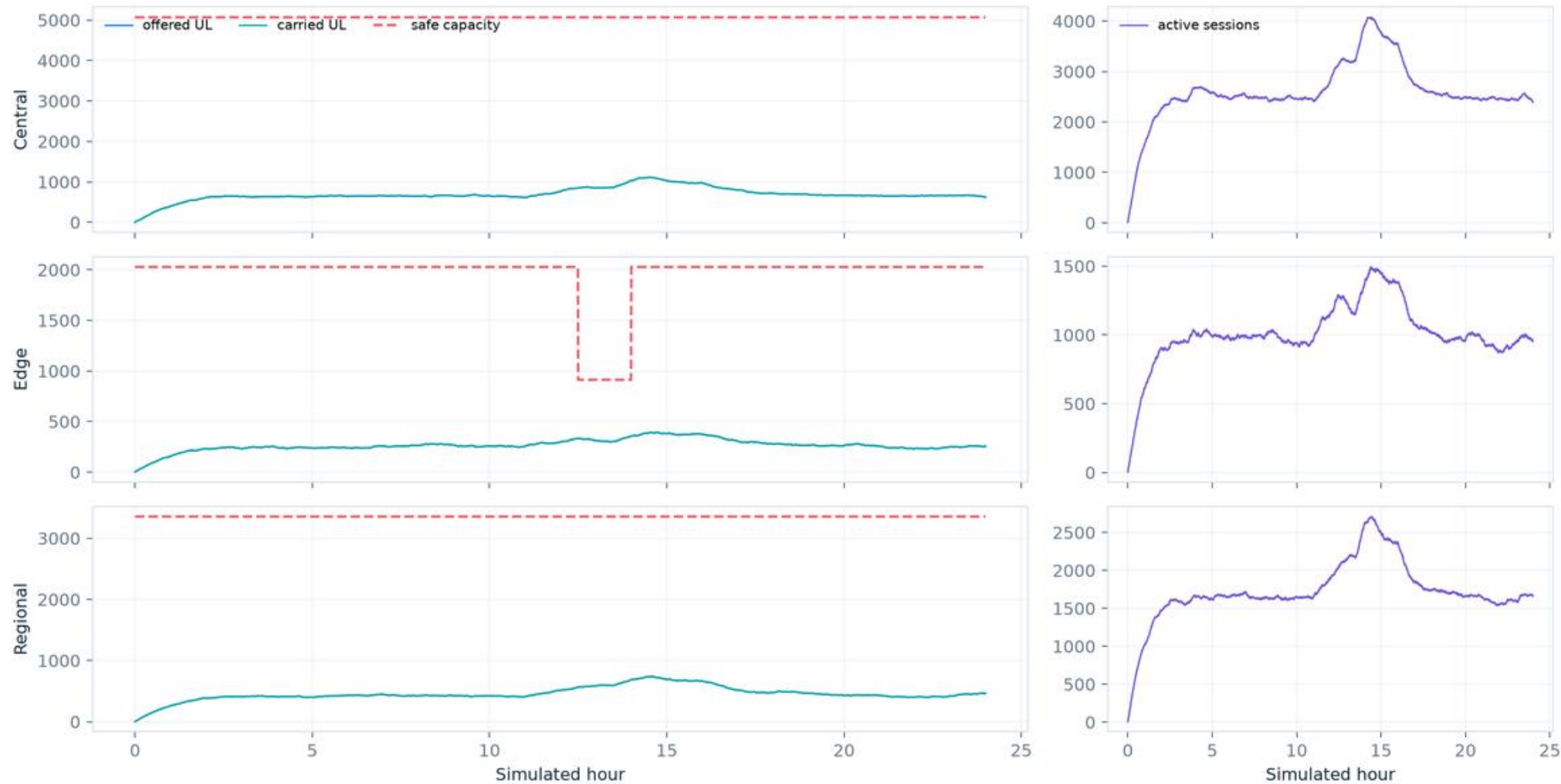
# Population is conserved while the traffic fingerprint changes by hour

## Population and traffic fingerprint



# A representative UPF day exposes edge, regional and central behavior

Representative edge / regional / central UPF day



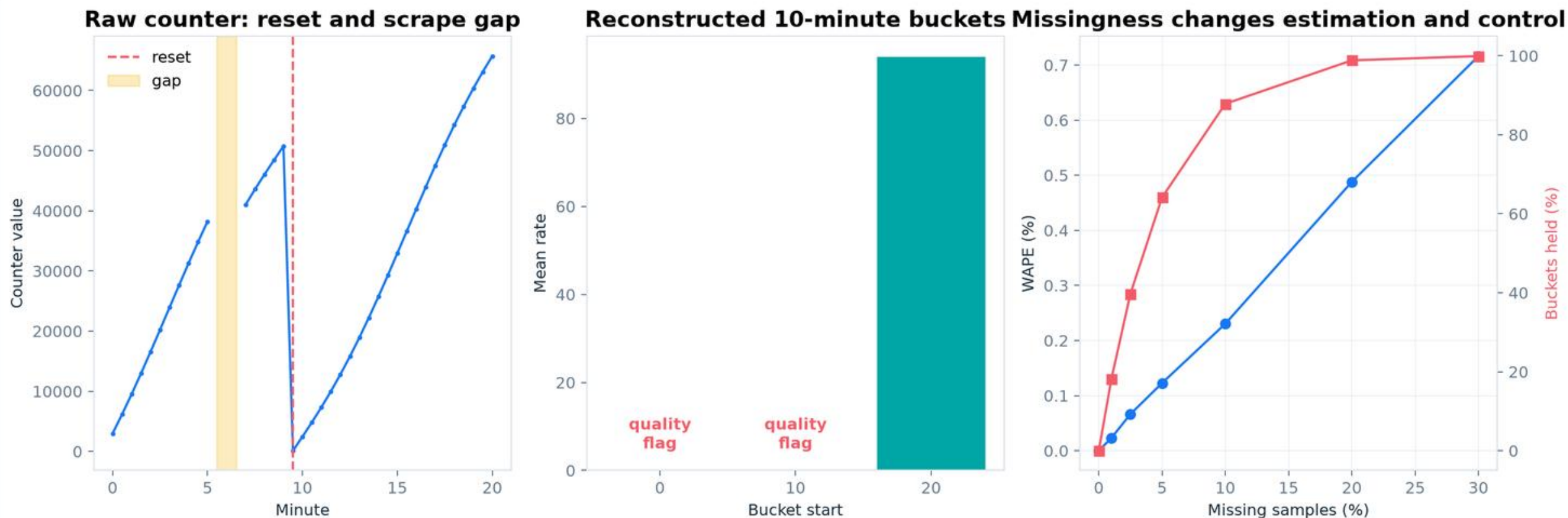


# Variable-rate accounting closes without reconstructing load from session counts

Directional accounting closes at machine precision



# Telemetry pathologies are visible and quality flags propagate



# Offline evidence plane: immutable inputs to frozen artifacts



- Every displayed result resolves to a source path and SHA-256 in one presentation manifest.
- Seed matrices, model/profile identity and source fingerprints remain separate per experiment.

# Online loop: observe → forecast → optimize → certify → steer → measure



- Policy is selected only after a bucket closes.
- Weights affect later arrivals; established sessions stay anchored.
- Fallback retains static or the last safe policy when state, telemetry or certificate quality is insufficient.



# MPC carries cohort state across 12 windows and proves the first move

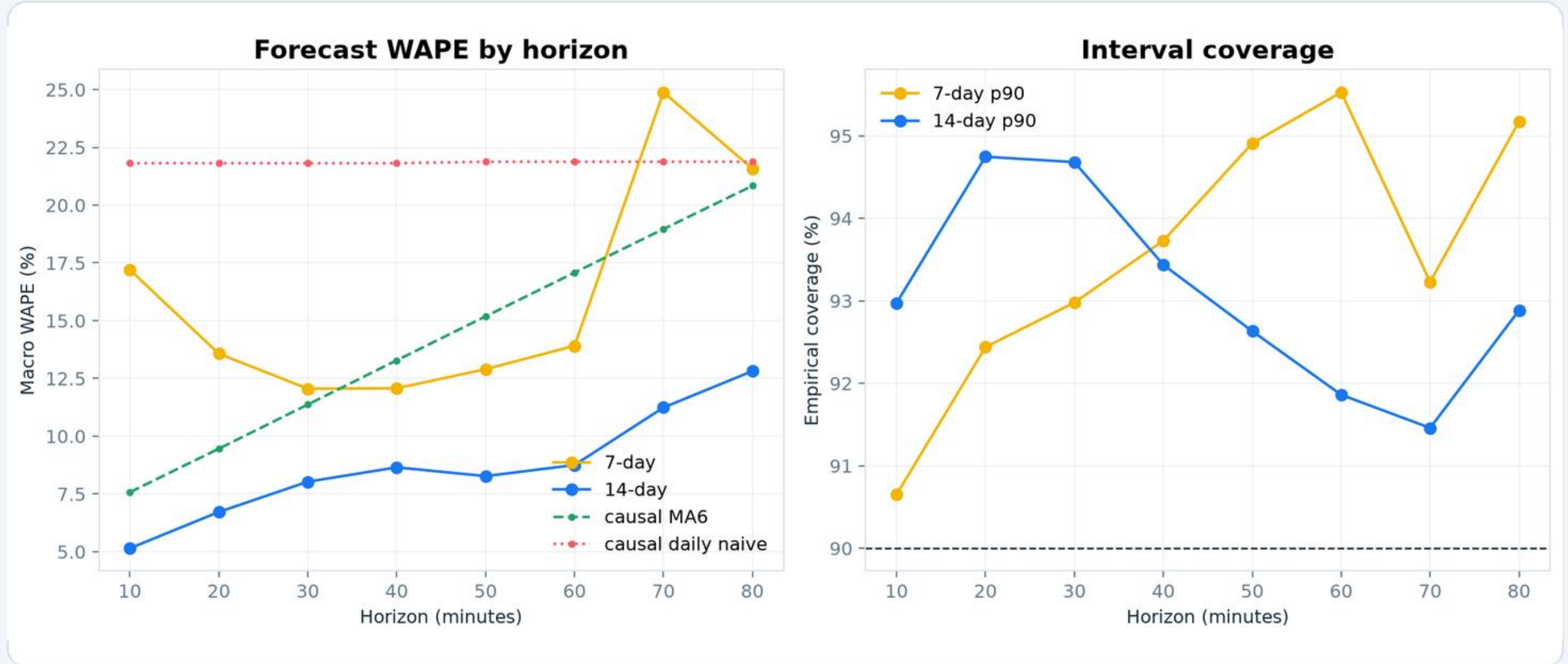
- State: active cohorts by group, UPF, rate bin and remaining lifetime.
- Action: normalized group→UPF weights for future sessions.
- Objective: directional overload + drops + terminal exposure + policy deviation.
- Constraints: UL/DL/session capacity, health, eligibility, locality and diversification.
- Certificate: compare MPC and static from the identical causal state; publish only a safe first action.

# The experiment journey includes the failures that changed the design



**The engineering conclusion came from negative results: one-window optimization concentrated persistent sessions, and forecast accuracy alone did not guarantee a better policy.**

# Forecast accuracy and interval coverage improve in the fourteen-day corpus



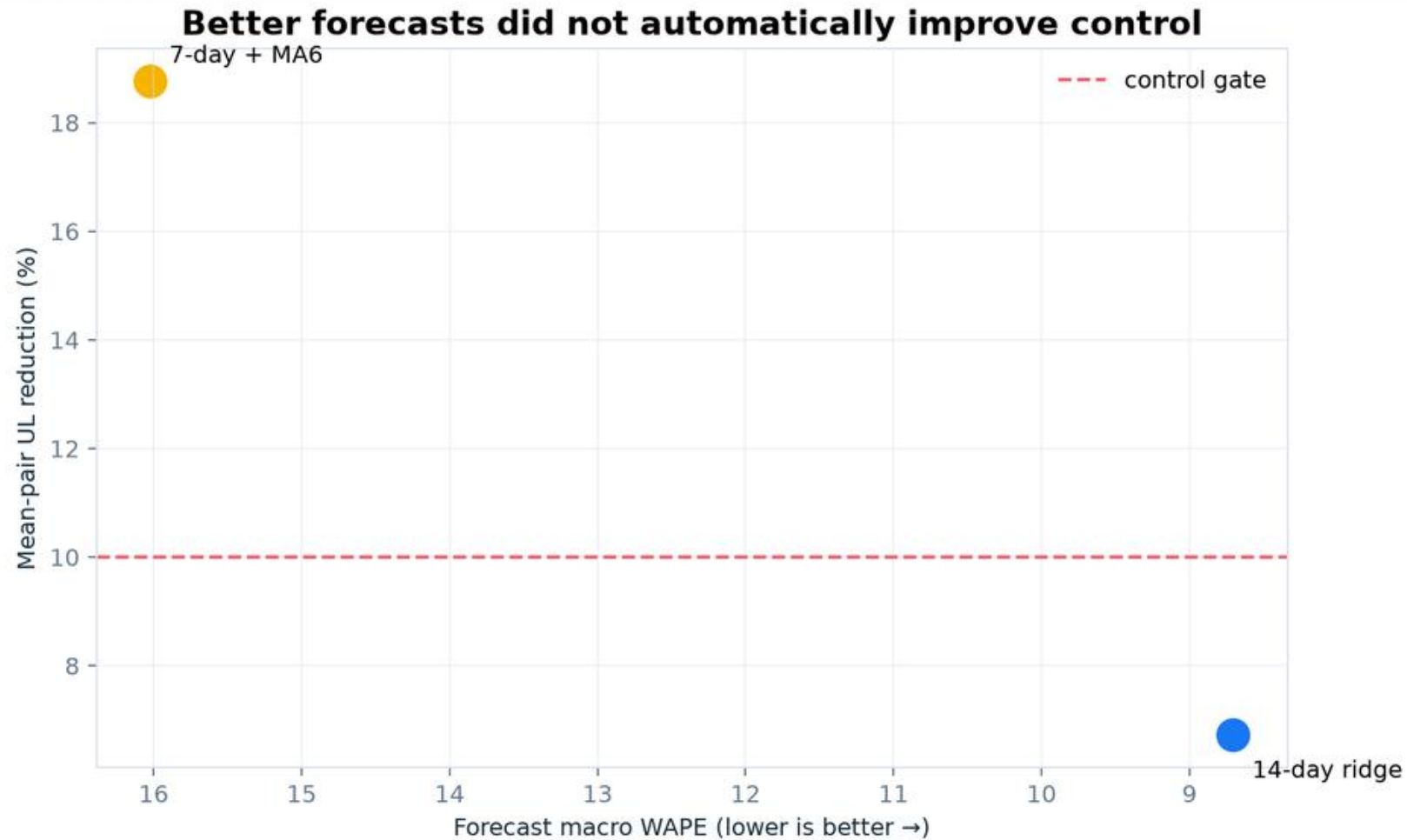


# Why the first optimizers failed: they optimized a window, not persistence

- One-window HiGHS reduced projected peak utilization but concentrated allocations across correlated groups.
- Many sessions lasted for hours, so an apparently good placement became tomorrow's residual overload.
- Static capacity weighting remained a strong baseline because it diversified continuously.
- The fix was structural: explicit cohort survival, terminal exposure and a same-state static certificate.

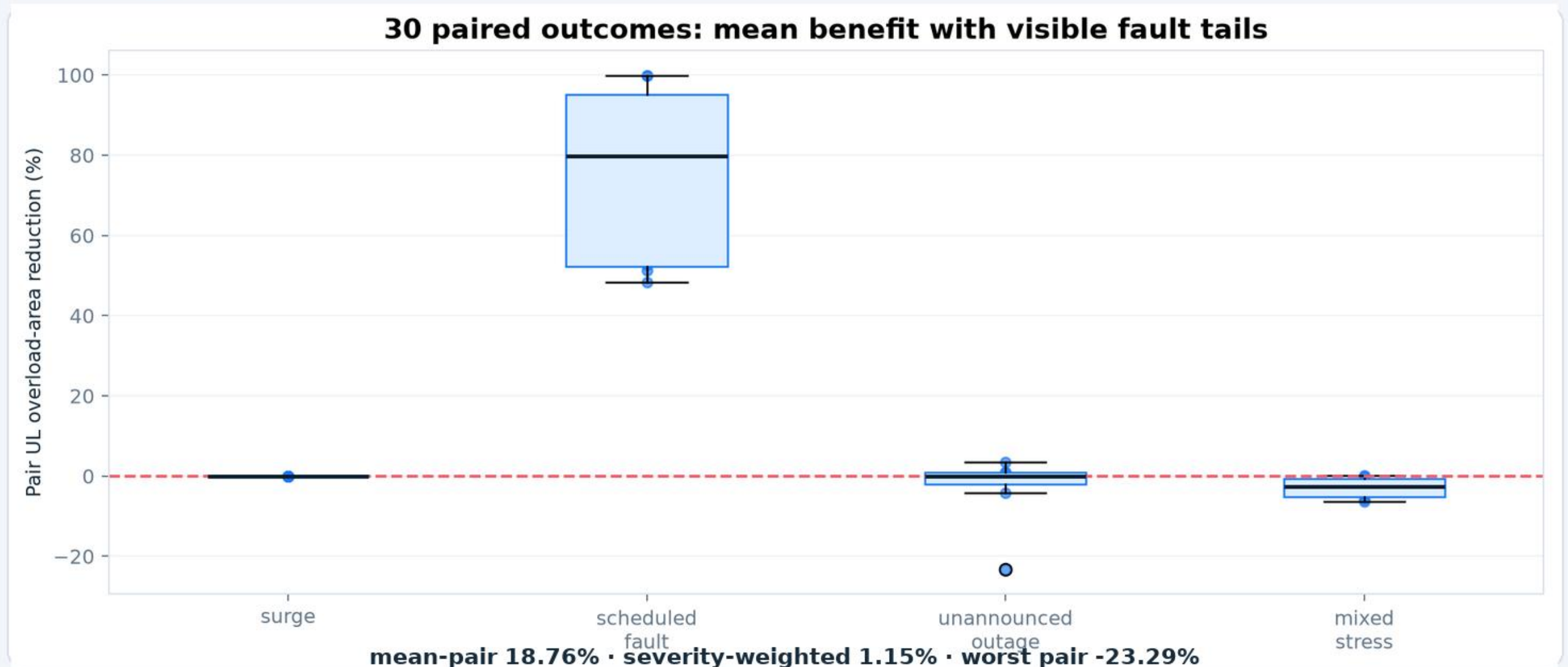


# Better forecasts did not automatically improve closed-loop control

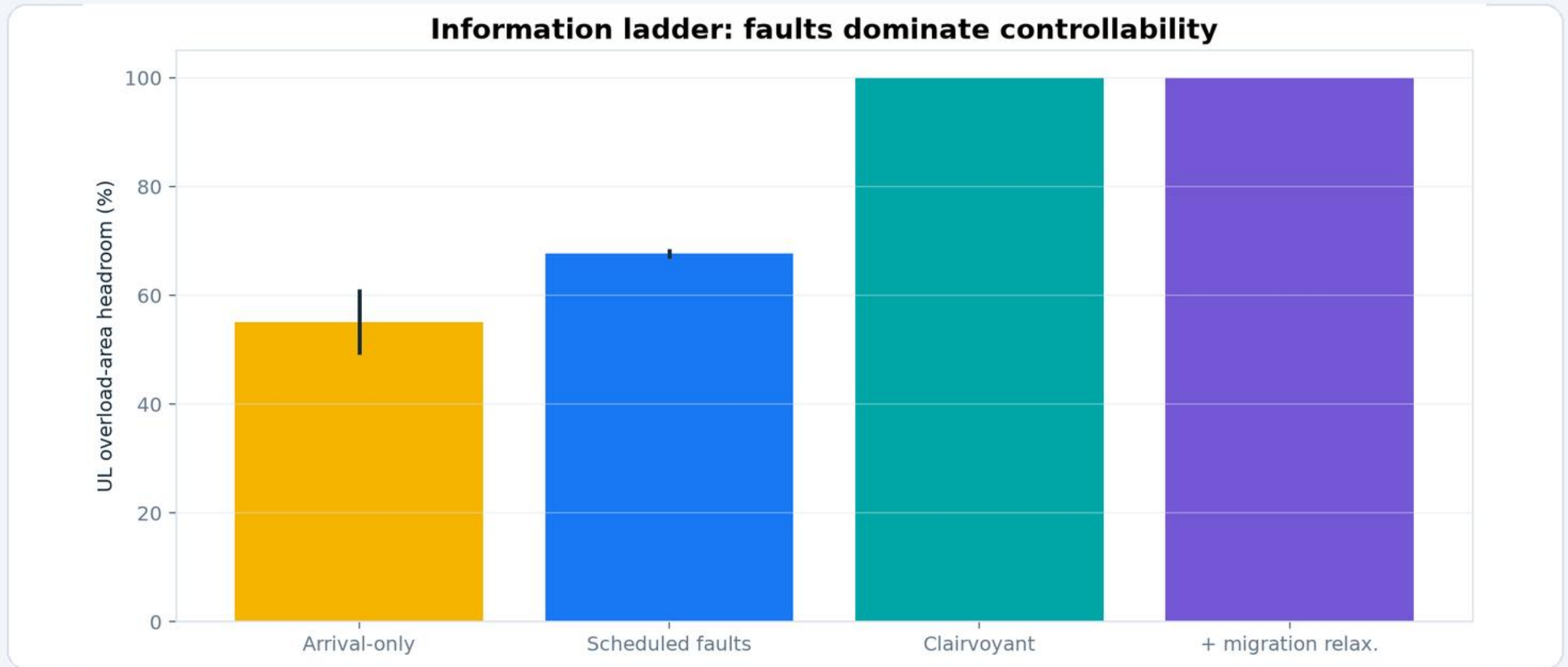


**14-day WAPE 8.70%; MPC mean-pair 6.72%; severity-weighted -3.60%.**

# The accepted national MA6 result passes on average—and exposes fault tails



# Oracle headroom isolates the causal fault-information gap



# HPC evidence: 8 workers selected; 2→4→12 chart remains withheld



**2 → 4 → 12 nodes**

**walltime/throughput/CPU/memory chart is withheld until every 2→4→12 pilot report passes**

EXTERNAL - PENDING



# Hybrid demonstration: live causality, then frozen national evidence

1

## Run reliable guided story

3-UPF causal runtime

2

## Expose the decision

forecast → certificate → weights

3

## Show realized effect

future-session placement and later overload

4

## Switch to frozen scale

24 UPFs · 96 groups · hashes

## Conclusion: proven synthetic control loop; C-DOT co-design is the next evidence step

- Proven here: causal new-session loop, deterministic v1/v2 simulation, hashed campaign evidence and same-state certification.
- Requested from C-DOT: UPF counters, topology/eligibility truth, capacity envelopes and a supported new-session selection hook.
- Proposed progression: advisory replay → shadow recommendations → bounded pilot with fail-closed rollback.

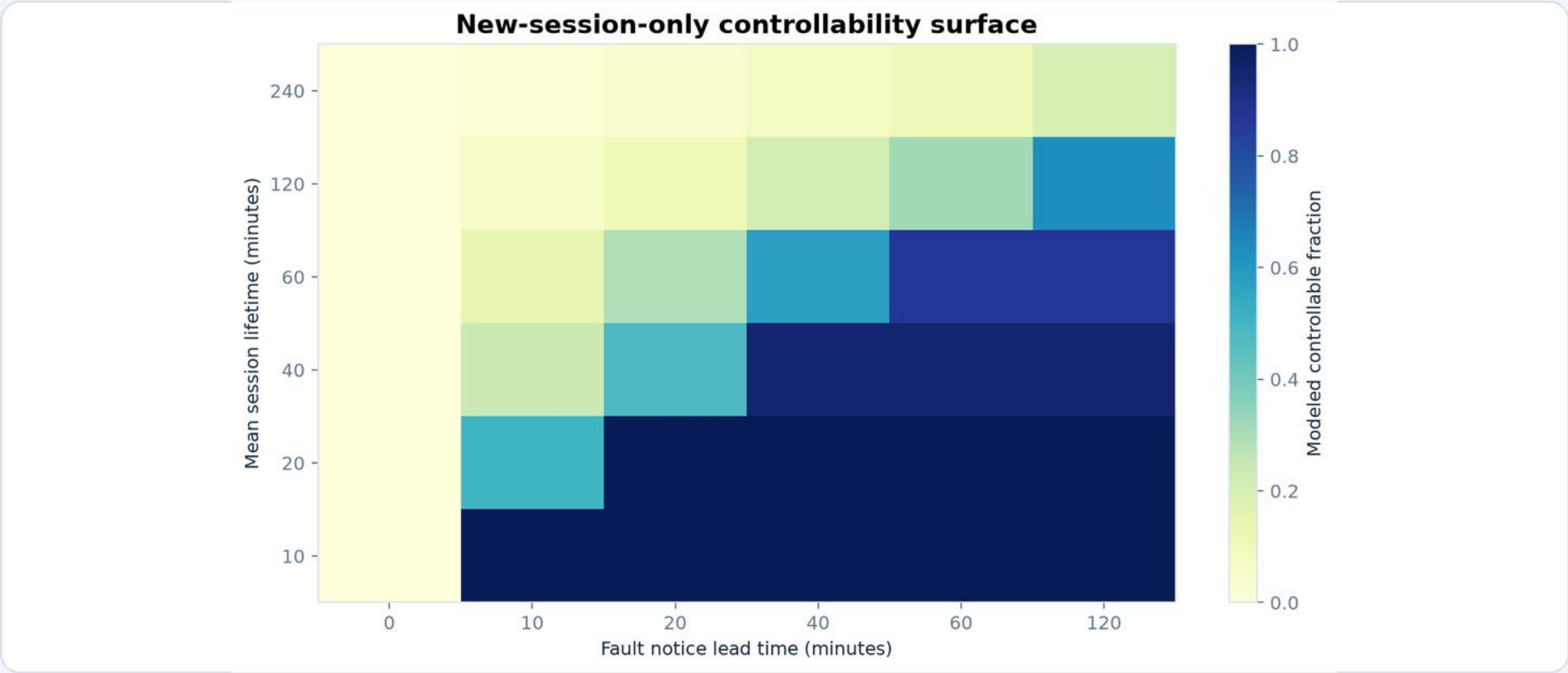
**No claim of production calibration, autonomous actuation, established-session migration or production readiness.**

## Appendix: reveal sequence and offline fallback



**The GIF, slide images, HTML and PDF are fully offline. Browser reconnect does not alter server-side realized history.**

# Appendix: controllability depends on notice and session lifetime





# Appendix: traceability is machine-verifiable

|                         |                                              |                             |
|-------------------------|----------------------------------------------|-----------------------------|
| demo_scenario           | demo_mpc_scenario.json                       | 919e003d4c4bde4b058e0475... |
| forecast_14d            | extreme-forecaster-14d-s20260818.freeze.json | 8c673d6a2a9e8ab926d1cbaf... |
| forecast_7d             | extreme-forecaster-7d-s20260817.freeze.json  | e8e3a2b50e5ff68f82871386... |
| forecast_baselines_base | baseline-evaluation.json                     | f8eed18b8d89373c528aa5af... |
| fourteen_day            | evaluation.json                              | f4f1c5b3c10908ea2f7d7566... |
| guided                  | cohort_mpc_full_campaign_evidence_v1.json    | 4c76e36eebe6f2245efb1c81... |
| ma6                     | evaluation.json                              | 923f52f7fbf96dc91cd7ce78... |
| oracle                  | oracle-bound-evaluation-s036001-s036002.json | 34d8b8b5d4f1b947e91d8327... |
| packing                 | stage1-report.json                           | 3732b7dab778b94cf642bb72... |
| performance             | traffic-v2-performance.json                  | 469fb2348ab07d2d946130ca... |

## Appendix: v2 controller pilot is gated and intentionally pending

FRESH PILOT

**16 pairs**

four seeds × four stress classes

ADVANCE THRESHOLD

**10%**

positive severity-weighted + no  
guardrail regression

CURRENT STATUS

**NOT RUN**

accepted v1 profile remains  
unchanged

**If realism changes controller behavior, that result will be presented as evidence—not used to overwrite the accepted v1 campaign.**